

HOSNAIN RAFI

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SUMMARY

Full-stack engineer with **3+ years of professional engineering experience** building scalable, real-time, and **AI-powered web applications** using MERN, Angular, TypeScript, and modern cloud technologies. Specialized in real-time systems (Socket.IO, WebRTC), agentic AI workflows, LLM vision quality gates, and high-throughput microservices. Proven ability to design and deliver production-ready applications with a strong focus on performance, user experience, and scalability. Committed to disciplined professional practices: proactive communication (**Hou-Ren-So**), continuous improvement (**Kaizen**), and **Quality-First** engineering standards.

EXPERIENCE

Software Engineer | Mortgage Magic Ltd March 2023 – Present
Remote for a London-based company | Angular, Node.js, Express, PostgreSQL, Redis, Docker, Playwright

- **Developed and shipped production-ready Angular features:** Built and maintained responsive frontend modules using Angular, TypeScript, HTML, and CSS, improving application usability and performance.
- **REST API & Microservices Integration:** Integrated frontend components with backend REST APIs, ensuring efficient data handling, schema validation, and seamless user experience.
- **High-Throughput Backend Architecture:** Architected decoupled services in Node.js and Express with PostgreSQL and Redis caching, reducing API response times by 35% on core pipeline workflows.
- **Automated Ingestion & Rate Scraping:** Developed automated rate-scraping engines and criteria ingestion pipelines, integrating intelligent data extraction to reduce manual advisor entry by 60%.
- **Bug Remediation & System Stability:** Identified and resolved critical bugs across frontend and backend layers, improving system reliability across releases.
- **Performance Optimization:** Contributed to UI optimization, reducing DOM render latency and enhancing responsiveness for high-traffic financial portals.
- **Quality Gates & Code Reviews:** Participated in disciplined peer code reviews and maintained Playwright end-to-end regression suites, ensuring clean, scalable, and well-documented codebases.
- **Cross-Functional Collaboration:** Collaborated with UI/UX designers, backend engineers, and product teams to deliver features aligned with business requirements.
- **Version Control & Team Workflows:** Managed version control, branching strategies, and CI/CD workflows using Git and GitHub.

SKILLS

AI & Machine Learning: Claude Vision API, OpenAI & Gemini APIs, Multi-Agent Swarms, Autonomous Task Orchestration, Tool / Function Calling, Automated UI Quality Gates (DesignGate), Prompt Engineering, RAG Architectures, Vector Embeddings

Languages: JavaScript (ES6+), TypeScript, C, C++, C# (.NET Core), Python, HTML5, CSS3, SQL

Frontend: React, TypeScript, Angular, Vue.js, Next.js, Redux Toolkit, Pinia, Tailwind CSS, Responsive Design, State Management, DOM Optimization

Backend: Node.js, Express, RESTful APIs, Microservices Architecture, WebSocket / Socket.IO, WebRTC, Redis Pub/Sub, gRPC

Databases: MongoDB, PostgreSQL, MySQL, Redis (In-Memory Cache), SQLite, Prisma ORM, Mongoose

DevOps: Docker, Git, GitHub, Bitbucket, Vercel, Netlify, Heroku, Linux (Ubuntu/CentOS), Web Scraping (BeautifulSoup/Puppeteer), Vite

Testing & QA: Playwright E2E Testing, Jest Unit Testing, Postman API Automation, Visual Regression AI Analysis, CI/CD Pipelines

PROJECTS

DesignGate — Automated UI Quality Gate with Vision LLM

[GitHub](#) | 2025

Tech: Playwright Headless, Claude Vision API, Node.js, TypeScript, Next.js, Tailwind CSS

- Built an automated QA quality gate eliminating subjective UI code reviews by combining automated headless browser evidence collection with multi-modal AI evaluation.
- **Automated Evidence Pipeline:** Captures synchronized desktop and mobile viewports, full-page snapshots, and DOM hierarchy using Playwright.
- **Vision LLM Quality Grading:** Integrates Claude Vision to evaluate design fidelity, layout drift, typography consistency, contrast ratios, and responsive anomalies against baseline specifications.
- **CI/CD Readiness:** Generates machine-readable JSON pass/fail verdicts and interactive visual audit reports, enabling zero-human-touch visual regression gating in deployment pipelines.

Agent-Swarm — Autonomous Multi-Agent CLI Orchestrator

[GitHub Repo](#) | 2025

Tech: Python, OpenAI / Claude APIs, Asyncio, Multi-Agent Architecture, CLI Tooling

- Engineered a distributed multi-agent swarm architecture for developer terminal assistants to execute complex, multi-stage software engineering workflows autonomously.
- **Hierarchical Agent Delegation:** Implemented specialized sub-agent roles (Planner, Coder, Reviewer, and Test Runner) communicating via structured message buses and shared context memory.
- **Self-Correction & Tool Execution:** Agents autonomously execute bash scripts, inspect source code trees, parse compiler/linter errors, and iterate on bug fixes until acceptance criteria are satisfied.

Urban Grill — Real-Time Operations & Restaurant Platform

[Live Site](#) | [GitHub Repo](#) | 2025

Tech: React, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS, Vercel

- Developed a full-stack restaurant operations platform with dual customer ordering and live kitchen administrative management.
- **Real-Time Kitchen Dispatch:** Engineered low-latency bidirectional WebSocket (Socket.IO) pipelines synchronizing orders instantly between customer mobile devices and kitchen display terminals.
- **Role-Based Operational Workflows:** Implemented JWT authentication and fine-grained access control across customer, kitchen staff, and executive admin portals with live inventory tracking.

EDUCATION

B.Sc. (Engg.) in Information and Communication Technology (ICT) | Mawlana Bhashani Science and Technology University

2019 – 2023

Tangail, Bangladesh

LANGUAGES & PROFESSIONAL PRINCIPLES

English: Business Level — Fluent in technical architecture discussions, sprint documentation, and global team collaboration.

Japanese: Basic / Conversational — JLPT N5 preparation in progress (daily study of technical terminology and conversation).

Work Principles: Active practitioner of Hou-Ren-So (proactive reporting, contact, and consultation), Kaizen (continuous iterative improvement), and Quality-First engineering standards.